

EQUATIONS , PROFIT AND LOSS, PERCENTAGES

Directions for questions 1 and 2: Select the correct alternative from the given choices.

If $2x - 3y = 1$ and $5x + 2y = 50$, what is the value of $(x - y)$?

☒ a) 3 (Your answer - ✓)

☐ b) 6

☐ c) 10

☐ d) 7

Directions for questions 1 and 2: Select the correct alternative from the given choices.

In an examination 35% of students passed in Maths, 48% passed in Science and 23% passed in both. What percent of the students failed in both the subjects?

☐ a) 50%

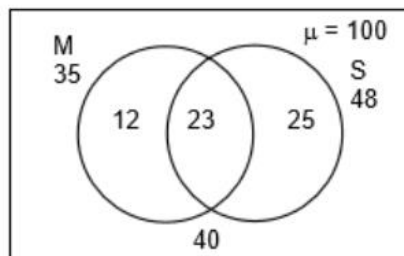
☐ b) 40% (Correct Answer - ✓)

☐ c) 60%

☐ d) 53%

Solution:

The data is shown in the Venn diagram below.



As 23% passed in both, 12% passed in only Maths, and 25% ($48 - 23$) passed in only Science. Therefore, the percentage of students who failed in the both subjects is $100 - (12 + 23 + 25) = 40$.
Choice (B)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

If a four digit number $7p6p$ is divisible by 11, what is the maximum number of values that p can take?

Your Answer: 3 ✖

Correct Answer: 1 ✔

Solution:

The number $7p6p$ is divisible by 11. Therefore, $E = 2p - (6 + 7)$ has to be a multiple of 11.
As $0 \leq 2p \leq 18$, the limits for E are $-13 \leq E \leq 5$. Also, E is an odd multiple of 11.
 $2p - 13 = -11 \Rightarrow p = 1$. This is the only value of p , for which $7p6p$ is a multiple of 11.
Ans : (1)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

The ratio of the number of currency notes of denominations Rs.10, Rs.20 and Rs.50 with Badal is 5 : 3 : 2. If he has Rs.840 with him, how many notes of denomination Rs.20 does Badal have?

Your Answer: 240 ✖

Correct Answer: 12 ✔

Solution:

The data is tabulated below.

Denomination	10	20	50
Ratio	5	3	2
Number	$5x$	$3x$	$2x$

$$\therefore 50x + 60x + 100x = 840 \Rightarrow x = 4$$

The number of 20 – rupee notes is $3x = 3(4) = 12$ Ans : (12)

Directions for question 5: Select the correct alternative from the given choices.

The manufacturing cost of a watch is Rs.180 and the transportation cost is Rs.500 for 100 watches. What will be the selling price of each watch if it is sold at 20% gain?

☐ a) Rs.222 (Correct Answer - ✓)

☒ b) Rs.216 (Your answer - ✗)

☐ c) Rs.221

☐ d) Rs.220

Solution:

The manufacturing cost of one watch is ₹180 and the transportation cost per watch is ₹5. The total cost is ₹185.
20% of this is 37.
∴ The watch has to be sold at ₹222.

Choice (A)

Directions for question 6: Type in your answer in the input box provided below the question.

What is the units digit in the expansion of $2^2 \times 3^3 \times 4^4$?

Your Answer: 2 ✗

Correct Answer: 8 ✓

Solution:

The units digits of 2^2 , 3^3 , 4^4 are 4, 7, 6 respectively.
∴ The units digit of the product is that of $4(7)(6)$, which is 8.

Ans : (8)

Directions for questions 7 and 8: Select the correct alternative from the given choices.

The mean proportional of the two positive numbers obtained by subtracting 55 and 22 from a positive number is 22. Find the number.

☐ a) 55

☐ b) 33

☒ c) 66 (Correct Answer - ✓)

☐ d) 77

Solution:

Let the number from which 55 and 22 are subtracted be x .

∴ The mean proportional of $x - 55$, $x - 22$ is 22

$$\text{i.e. } (x - 55)(x - 22) = 22^2$$

$$\Rightarrow x^2 - 77x + 11(110) = 11(44)$$

$$\Rightarrow x^2 - 77x + 11(66) = 0$$

$$\Rightarrow (x - 66)(x - 11) = 0.$$

As $x - 55 > 0$, x has to be 66.

Choice (C)

Directions for questions 7 and 8: Select the correct alternative from the given choices.

25 men can construct a wall in 4 days. If the work is to be completed in 10 days, then how many men can take leave?

☒ a) 10 (Your answer - ✗)

☐ b) 15 (Correct Answer - ✓)

☐ c) 5

☐ d) 12

Solution:

25 men can complete the work in 4 days. The work is 100 mandays. If the work has to be completed in 10 days, only 10 men are needed, i.e. 15 of the 25 men are not needed.
Choice (B)

Directions for question 9: Type in your answer in the input box provided below the question.

Srinivas and Raju rent a pasture for 8 months. Srinivas feeds his 60 oxen on the pasture for 5 months. How many oxen can Raju feed for the remaining 3 months, if he pays one and a half times as much rent as Srinivas?

Your Answer: 100 ✖

Correct Answer: 150 ✔

Solution:

The data is tabulated below,

	Srinivas	Raju
No of oxen	60	
Period	5	3
Rentpaid (Ratio)	2	3

If the rent had been shared equally, Raju would be entitled to feed 100 oxen [as $60(5) = 100(3)$]. But as he pays one and a half times the rent paid by Srinivas, he is entitled to feed 150 oxen.
Ans : (150)

Directions for question 10: Select the correct alternative from the given choices.

A man earns twice as much as his wife does. His expenditure is 80% of his wife's income and he saves Rs.1,800. What is his wife's income?

☐ a) Rs.1,500 (Correct Answer - ✔)

☐ b) Rs.1,200

☐ c) Rs.2,000

☐ d) Rs.1,750

Solution:

The data is tabulated below.

	Man	Wife
Savings	1800	
Expenditure	4x	
Income	10x	5x

If the wife's income is $5x$, the man's would be $10x$ and his expenditure would be $4x$, i.e. savings would be $6x$.

$6x = 1800 \Rightarrow x = 300$. The wife's income is $5x = ₹1500$.

Choice (A)

Directions for question 11: Type in your answer in the input box provided below the question.

The sum, difference and the product of two numbers are in the ratio $3 : 1 : 20$. Find the product of the numbers.

Your Answer: Unattempted

Correct Answer: 200 ✓

Solution:

Let the two number be a, b .

$$\therefore a + b : a - b : ab = 3 : 1 : 20$$

$$\Rightarrow \frac{a+b}{a-b} = \frac{3}{1} \Rightarrow \frac{a}{b} = \frac{3+1}{3-1} = 2$$

$$\therefore a + b : a - b : ab = 3b : b : 2b^2 = 3 : 1 : 2b$$

As $3 : 1 : 20 = 3 : 1 : 2b$, it follows that $b = 10$.

$$\therefore a = 20 \text{ and } ab = 200.$$

Ans : (200)

Directions for questions 12 and 13: Select the correct alternative from the given choices.

The LCM and the HCF of two numbers p and q , are x and y respectively. If $x = p$ and $p > q$, q is equal to _____.

- ☐ a) x/p
- ☐ b) p/q
- ☒ c) y (Correct Answer - ✓)
- ☐ d) y/p

Solution:

The greater number, p is equal to the LCM, x , the smaller one has to be the HCF y , i.e. $q = y$.
Choice (C)

After getting three equal successive percentage discounts over a marked price of Rs.1000, a customer has to pay Rs.729 for an article. What is the rate of each of the successive discounts?

- ☒ a) 10% (Your answer - ✓)
- ☐ b) 20%
- ☐ c) 30%
- ☐ d) 40%

Direction for question 14: Type in your answer in the input box provided below the question.

How many prime numbers are there between the smallest and the largest two digit prime numbers less than 50?

Your Answer: 11 ✗

Correct Answer: 9 ✓

Solution:

The smallest and the largest two – digit prime numbers less than 50 are 11 and 47 respectively and there are 9 prime numbers between them viz. 13, 17, 19, 23, 29, 31, 37, 41 and 43.
Ans : (9)

Direction for question 15: Select the correct alternative from the given choices.

If one is added to both the numerator and the denominator, a fraction becomes $\frac{1}{2}$.
Instead if one is subtracted from both the numerator and the denominator, the fraction becomes $\frac{3}{7}$. What is the fraction?

☐ a) $\frac{6}{11}$

☒ b) $\frac{7}{15}$ (Your answer - ✓)

☐ c) $\frac{8}{17}$

☐ d) $\frac{5}{9}$